

Thermal comfort in the humid tropics: Field experiments in air conditioned and naturally ventilated buildings in Singapore

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Abstract. Thermal comfort field experiments were conducted in Singapore in both naturally ventilated high-rise residential buildings and air conditioned office buildings. Each of the 818 questionnaire responses was made simultaneously with a detailed set of indoor climatic measurements, and estimates of clothing insulation and metabolic rate. Results for the air conditioned sample indicated that office buildings were overcooled, causing up to one-third of their occupants to experience cool thermal comfort sensations. These observations in air conditioned buildings were broadly consistent with the ISO, ASHRAE and Singapore indoor climatic standards. Indoor climates of the naturally ventilated apartments during the day and early evening were on average three degrees warmer than the ISO comfort standard prescriptions, but caused much less thermal discomfort than expected. Discrepancies between thermal comfort responses in apartment blocks and office buildings are discussed in terms of contemporary perceptual theory.

Key words: Thermal comfort – Field study – Indoor climate – Energy conservation – Perceptual theory

Introduction

Thermal comfort research can be performed in either a climate chamber or in field settings (typically buildings). The former methodology permits an independent environmental variable to be manipulated directly whilst isolating the dependent variable, comfort level, from extraneous influences. While this controlled research design has permitted the relative importance and interactions of several independent variables to be disentangled, unfortunately this reduces thermal comfort to a simplistic stimulus-response system (McIntyre 1982). Environ-

mental psychologists have long contended that laboratory studies represent crude oversimplifications of person-environment interactions and consequently have doubted their relevance to the solution of practical problems of the built environment (Proshansky 1972; Russell and Ward 1982). However, since researchers from engineering and physical science backgrounds rather than psychology have dominated thermal comfort research, climate chambers have remained the principal research tool for supplying professionals of the built environment with data on human thermal requirements (e.g. ISO 1984; ASHRAE 1981).

Field studies of thermal comfort on the other hand are characterized by greater external validity than the laboratory-based methodology. Field-based researchers recognize the person-environment system as an integral unit in which sensation and perception are influenced by the thermal environment, which in turn is modified by behaviour in a self-regulating manner (Nicol and Humphreys 1973). Being conducted in naturalistic settings, field studies also avoid the artificiality of the climate chamber (McIntyre 1982), thereby preserving the phenomenal integrity of the environment under study (Stokols 1977). Furthermore, field studies are typically based on large numbers of respondents, thus further enhancing external validity. In comparison, laboratory studies are usually constrained to small sample sizes on account of the requirement to pay hourly rates for subjects' participation.

Since field studies are usually conducted to answer specific questions about a single building, their results rarely have general relevance to comfort theoreticians. Whilst individually they may not be so significant, viewed collectively the extensive body of field studies have suggested some interesting hypotheses about the relationship between the temperatures that people find comfortable and prevailing levels of environmental warmth. Comprehensive reviews and statistical analyses of the results of over 50 individual field studies on thermal comfort from various global climatic regimes have

been published by Humphreys (1976, 1981) and Auliciems (1981, 1983). Both researchers reported strong positive correlations between the observed comfort temperature and the mean temperatures prevailing both indoors and outdoors during the field studies. This relationship is widely considered to be inconsistent with thermal comfort models derived from climate chamber research (Auliciems 1981; Humphreys 1976; McIntyre 1982).

Obviously a thorough understanding of the comfort problem requires contributions from both field- and laboratory-based research. A third methodological approach, that of the field experiment, has recently been used to test critically the inconsistencies of the other two (de Dear and Auliciems 1985; Schiller et al. 1989; Busch 1990). It is essentially the same as a field study in that the research is conducted in actual buildings with 'real' occupants as opposed to paid subjects. The distinguishing feature of the method is that all the environmental and behavioural variables known from climate chamber experiments to influence thermal comfort are measured in situ. In contrast with earlier field studies which typically measured only air temperature and humidity, a complete set of measurements in field experiments permits the prediction of body-environment heat balances for each respondent. The basic idea of this approach is that, if several independent variables cannot be controlled in the field, they can at least be measured so that subsequent data analyses can partial out their effects on comfort levels. Despite this additional attention given to environmental measurement, the results of field experiments have not always supported those of the climate chamber method (de Dear and Auliciems 1985; Schiller et al. 1989; Busch 1990). Three related postulates about the general nature of perception may offer an explanation for the discrepancies (Helson 1971; Ittelson 1973; Auliciems 1981; Russel and Ward 1982):

- (1) Perception is not exclusively determined by environmental stimulus or physiological responses.
- (2) Perception is not a discrete psychological process and is indistinguishable from memory and cognition. In this context the environmental expectations of a person come into play.
- (3) Perception is relevant to and appropriate for the environmental context in which it occurs.

Aims of the study

The current paper sets out to furnish new thermal comfort data based on field experiments in the humid tropics, which to date are underrepresented in the comfort literature (Humphreys 1981; Auliciems and de Dear 1986; Busch 1990). Two separate studies were conducted in Singapore, one in air conditioned offices and the other in naturally ventilated high-rise residential buildings. Comparisons between these two studies are made and the findings are then interpreted in relation to contemporary comfort theory and standards (ISO 1984; ASHRAE 1981) as well as the postulates of perception outlined above.

Location of the study

The climate of Singapore

Situated at latitude 1°21'N, the island of Singapore experiences a climate with uniformly high temperatures, high humidity, and abundant rainfall averaging 2381 mm per annum. The thermal uniformity is emphasized by the observation that the climatological mean monthly temperature varies by only 1.1 K from the mean annual value of 26.6° C. Diurnally there is also little variation, with the average daily range in temperature being 7 K. High sea surface temperatures in the adjacent South China Sea and Straits of Malacca cause the mean annual relative humidity to be 84% with typical daily maxima approaching saturation in the cooler early mornings (Singapore Meteorological Service 1987).

The limited seasonality that does exist is attributable to monsoonal shifts in prevailing wind directions and the attendant changes in cloud cover, rainfall, and solar radiation. Figure 1 depicts Singapore's thermal comfort contour surface by hour and month, based on the Standard Effective Temperature index (Gagge et al. 1986) applied to a lightly clad (0.5 clo) person, standing in an open field and facing the sun for the entire year of 1988 meteorological data (de Dear 1989). In the figure can be seen a reduction in thermal discomfort from November through March during the northeast monsoon, which is when wind velocity and cloud cover reach their seasonal maxima.

The air conditioned buildings were surveyed in the months of July and August 1986, during which the average air temperature was 27.4° C (ranging from an average daily minimum of 25.0° to an average maximum of 31.1° C), and the average relative humidity was

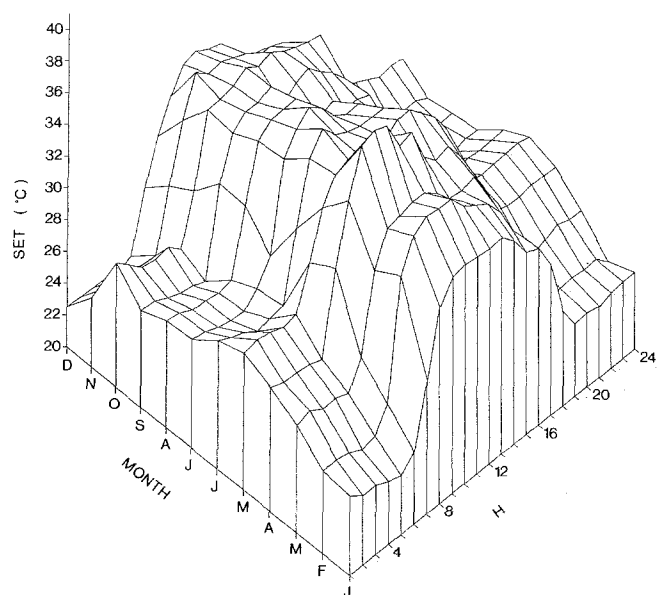


Fig. 1. Mean Standard Effective Temperatures (SET) by hour and month for 1988 in Singapore. SET was based on calculations for a subject dressed in 0.5 clo and standing in an open field for a full year of meteorological data (after de Dear 1989)

83.6%. The naturally ventilated buildings were surveyed a year later in August 1987 when the outdoor temperatures and humidities were comparable to the 1986 values.

The buildings surveyed

By 1987, the percentage of Singapore's population living in public housing was 85% (HDB 1987), most of which was concentrated in several high-rise residential clusters known locally as 'new towns'. This has resulted in population densities within the residential areas of the dozen or so new towns being as high as 100 000 persons per km² (HDB 1987). The four naturally ventilated buildings used in this study were selected from four representative new towns. In Singapore the most commonly used residential climate controls are ceiling and floor standing fans; Wong and Yeh (1985) found them in 85% of HDB apartments in 1981.

The twelve air conditioned office buildings used in the present study were selected from both the public and private sectors and were located primarily in the central business district of Singapore. All were high-rise buildings less than 5 years old and with their internal spaces typically landscaped and of open plan design. Air conditioning was provided from centrally controlled plant in all buildings.

Materials and methods

The respondents. Five hundred and eighty-three respondents from 214 households were interviewed in naturally ventilated buildings and 235 respondents were interviewed in air conditioned office buildings. All were either long-term residents of Singapore or had been born in the country. Basic demographic data on the two samples are summarized in Table 1. Senior personnel were not as accessible to the researchers as the younger staff in the office study, hence the relatively young sample in the air conditioned part of the study.

Indoor climatic measurements. Four atmospheric parameters, ambient air temperature, mean radiant temperature, humidity and air velocity, were measured simultaneously whilst the questionnaire was being administered. An Assmann aspirated psychrometer was used to measure indoor dry- and wet-bulb temperatures from which relative humidity was derived. Mean radiant temperature was assessed using a 150-mm-diameter globe thermometer. Wet-, dry- and globe thermometers were all calibrated with resolutions of 0.1 K. Globe temperature was converted to mean radiant temperature by calculating the convective and radiative heat balance of the globe. By making the simplifying assumption that convective

and radiative heat transfers from the human body were of equal significance, the single temperature for sensible heat loss known as operative temperature was estimated as the arithmetic mean of air and mean radiant temperatures. This index will be primarily used in the remainder of the paper. Indoor air velocities were measured by a Kanomax hot-wire anemometer (model 24-6111). All indoor climatic measurements were taken at a single height of 0.8 m above the floor and within a 1 m radius of the seated respondent.

Questionnaire. Clothing ensemble insulation was estimated by means of garment checklists compiled from various thermal manikin studies (Olesen and Nielsen 1983; McCullough et al. 1985). These checklists, one for each sex, yielded intrinsic ensemble insulation estimates by simply summing the individual garment insulation values in clo units (1 clo = 0.155 m² K per W) and multiplying by 0.82 (ISO 1984). Metabolic heat production in W/m² was assessed by means of an activity/behaviour checklist compiled from published tables of data (ASHRAE 1981; ISO 1984).

Apart from basic demographic items, the questionnaire also asked for thermal comfort responses. Respondents were asked to answer the following question by marking the standard seven-point scale (hot = +3; warm = +2; slightly warm = +1; neutral or just right = 0; slightly cool = -1; cool = -2; cold = -3) (McIntyre 1978): How does the temperature feel at this moment? Does the room feel cool, warm, or neutral (just right)?

Results

Indoor climates

Summary statistics for the 583 sets of indoor climatic measurements inside the naturally ventilated flats are given in Table 2. The mean radiant temperature was about half a degree warmer than the mean air temperature of 29.4° C, probably due to the timing of most interviews in the afternoon and evening hours. Consequently the mean operative temperature was marginally warmer than the air, at 29.6° C, and quite uniform with a standard deviation of only 1.2 K. Relative humidities were uniformly high throughout, with a mean of 74%. Indoor air velocities were light for naturally ventilated rooms, with a mean of 0.22 m/s. While fans were observed in the overwhelming majority of flats, they were rarely in use during the interviews.

The mean radiant temperatures measured in air conditioned buildings were on average more than one degree warmer than air temperature, probably resulting from high solar loads in Singapore combined with low thermal mass and large areas of glazing in the office buildings surveyed. The average operative temperature recorded in air conditioned buildings was 23.5° C. Air velocities

Table 1. Demographic data for the samples of respondents

		Age group (years)				Total
		17-20	21-40	41-60	> 60	
Natural ventilation	Male	37	143	70	37	287
	Female	50	98	95	53	296
	Total	87	241	165	90	583
Air conditioned	Male	0	69	22	0	91
	Female	18	118	8	0	144
	Total	18	187	30	0	235

Table 2. Summary of the indoor micro-climatic data

		Naturally ventilated ^a				Air conditioned ^b			
		Mean	SD	Max.	Min.	Mean	SD	Max.	Min.
Air temperature	(°C)	29.4	1.23	31.9	26.0	22.9	1.33	26.8	18.3
Relative humidity	(%)	73.5	6.6	97.8	57.9	55.5	7.6	74.1	35.6
Mean radiant temperature	(°C)	29.8	1.19	31.9	26.8	24.1	1.14	28.8	19.7
Operative temperature	(°C)	29.6	1.20	31.7	26.5	23.5	1.20	27.5	19.0
Air velocity	(m/s)	0.22	0.12	0.58	0.05	0.11	0.10	0.65	0.01

^a $n = 583$; ^b $n = 235$ **Table 3.** Summary of metabolic and clothing data

		Naturally ventilated ^a				Air conditioned ^b			
		Mean	SD	Max.	Min.	Mean	SD	Max.	Min.
Ensemble insulation	(clo)	0.26	0.09	0.53	0.12	0.44	0.10	0.67	0.29
Metabolic heat	(W/m ²)	69.6	17.8	165.0	45.3	67.4	12.4	116.0	58.0

^a $n = 583$; ^b $n = 235$ **Table 4.** Thermal comfort votes and operative temperature in naturally ventilated buildings

Operative temperature (°C)	Mean vote	-3	-2	-1	0	+1	+2	+3	Totals
26.6–27.5	-0.7	0	4	17	4	6	0	0	31
27.6–28.5	-0.1	0	5	34	34	21	4	0	98
28.6–29.5	0.3	0	3	17	51	34	5	1	111
29.6–30.5	0.8	0	1	20	76	80	36	7	220
30.6–31.5	1.7	0	0	1	7	33	46	13	100
31.6–32.5	2.0	0	0	0	0	4	14	5	23
	+0.66	0	13	89	172	178	105	26	583

were low but typical for air conditioning (ASHRAE 1981), with a mean value of 0.11 m/s, while the mean indoor relative humidity of 56% was considerably lower than outdoor values.

Behavioral variables

Statistical summaries of metabolic rates and clothing insulation values estimated for the occupants of both the air conditioned and naturally ventilated buildings are shown in Table 3. In the naturally ventilated apartments, the mean clothing insulation value of 0.26 clo reflects the casual dress codes of Singaporeans in their homes, with the typical male ensemble consisting of shorts and t-shirt, while the typical female ensemble consisted of a light skirt and blouse. Dress codes were more formal in air conditioned offices, with a mean insulation value of 0.44 clo which is comparable to typical office attire in summer in the US (ASHRAE 1981). For men this typically consisted of a light short-sleeve shirt and long trousers with shoes, while for women, the typical office attire comprised a half slip, light knee-length skirt, a light short-sleeve blouse and shoes/sandals. The mean metabolic rates estimated in both surveys were approximately equal at about 69 W/m² or 1.2 met. In the office buildings, the respondents were mainly desk-bound. The

apartment respondents were also mainly sedentary for the hour just before being interviewed.

Thermal comfort responses

In Tables 4 and 5, respectively, subjective assessments of the indoor climates in naturally ventilated and air conditioned buildings have been cross-tabulated against operative temperature. In naturally ventilated apartment blocks, the mean comfort vote of 583 respondents was +0.66 which corresponds approximately to half-way between 'just right' and 'slightly warm' on the seven-point scale used. Slightly more than half of all comfort votes in the naturally ventilated buildings were warmer than neutral. The mean thermal comfort vote recorded in air conditioned buildings was just on the cool side of 'just right' at -0.34. Approximately one-third of all votes recorded in the air conditioned buildings were cooler than neutral.

Thermal neutralities were calculated from thermal comfort and operative temperature cross-tabulations using the probit regression technique (Finney 1971; Balantyne et al. 1977). The term 'neutrality' is used here to denote the operative temperature that caused 50% of respondents to vote on the cool half of the seven-point scale, while the remaining 50% voted on the warm side.

Table 5. Thermal comfort votes and operative temperature in air conditioned buildings

Operative temperature (°C)	Mean vote	-3	-2	-1	0	+1	+2	+3	Totals
18.6-19.5	-2.0	0	1	0	0	0	0	0	1
19.6-20.5	-2.2	3	2	0	1	0	0	0	6
20.6-21.5	-2.0	3	3	1	1	0	0	0	8
21.6-22.5	-0.7	2	3	4	11	1	0	0	21
22.6-23.5	-0.4	6	6	20	38	11	3	0	84
23.6-24.5	-0.2	3	5	10	34	10	3	1	66
24.6-25.5	0.1	0	2	2	26	2	3	0	35
25.6-26.5	0.8	0	0	0	2	3	1	0	6
26.6-27.5	1.0	0	0	0	0	3	0	0	3
	-0.34	17	22	37	113	30	10	1	230

Expressed differently, thermal neutrality is the operative temperature most likely to elicit a thermal comfort vote of 'neutral' or 'just right'. The result for the occupants of naturally ventilated buildings was 28.5° C operative temperature (95% fiducial limits 28.2° to 28.8° C). In air conditioned buildings the thermal neutrality was estimated to be 24.2° C operative temperature (95% fiducial limits 23.6° to 25.1° C). The smaller sample size of the air conditioned survey, particularly in warmer temperatures, accounts for the wide fiducial limits (1.5 K) on that survey's neutrality estimate.

Discussion

Indoor climatic and thermal comfort standards represent practical guidelines for HVAC (Heating, Ventilation and Air Conditioning) engineers. Comparing air conditioning practices in Singapore with the relevant standards, the mean indoor climate in this sample of office buildings, consisting of 23.5° C operative temperature and 55% relative humidity (RH), falls near the cool limit of the local standard's 23°-25° C comfort range at 60% RH (SISIR 1983), which was directly based on the summer comfort zone prescribed in the US air conditioning standard (ASHRAE 1981). The observed mean conditions were also cooler (circa 1.5 K) than the international thermal comfort standard's (ISO 1984) recommendations of 25° C and 60% RH.

The number of thermal comfort field experiments carried out previously in tropical air conditioned buildings is small (Humphreys 1981), but Auliciems and de Dears' (1986) Darwin study in Australia's tropical north and Busch's (1990) study in Bangkok, Thailand are directly comparable since a common methodology was used. The neutrality observed in Darwin office buildings ranged from 23.9° to 24.2° C, depending on season. For all intents and purposes however, these Darwin results can be regarded as identical to the present Singapore result. Busch (1990) found that Thai office workers in air conditioned buildings in Bangkok had a neutrality of 24.5° C which also closely agrees with the Singapore and Darwin results. Therefore Darwin's largely European office population seems to have identical air conditioning requirements to those of the office populations in Southeast Asia.

Obviously the indoor climate observed in naturally

ventilated apartments in Singapore was considerably hotter and more humid than that found in air conditioned buildings. A mean operative temperature of 29.6° C and a mean relative humidity of 74% were observed in the apartments. The indoor climate/comfort standards discussed above indicate these conditions to be well beyond the comfort range (ISO 1984; ASHRAE 1981; SISIR 1983). For example, the warmest operative temperature at 70% RH recommended in the US standard as being acceptable to at least 80% of building occupants in summer conditions was only 22.5° C (ASHRAE 1981), which was 7 K cooler than the average conditions observed in the present naturally ventilated buildings. The ISO (1984) standard was more realistic, but still the recommended operative temperature for the average clothing, metabolic, air velocity and humidity conditions was three degrees cooler than the mean of 29.6° C actually observed in these buildings.

Busch's (1990) field experiment in naturally ventilated office buildings in Bangkok indicated a neutrality of 28.5° C ET* (effective temperature) which, even when converted back into operative temperature, can be regarded as being in good agreement with the current Singapore result of 28.5° C. Both these recent findings are considerably warmer than the much earlier Singapore survey neutralities of 26.7° and 27.2° C obtained by Ellis (1953) and Webb (1959) respectively, but an explanation of the discrepancy is not possible since the earlier researchers failed to record all of the indoor climatic variables that are now known to affect the body's heat balance.

Apart from forming the analytic basis of indoor climate and comfort standards, mathematical models such as Fanger's (1970) Predicted Mean Vote (PMV) can be used to calibrate comfort responses, thereby facilitating standardized comparisons between different field experiments. By solving the PMV equation for operative temperature rather than comfort vote, the model's predicted neutral temperature under the mean indoor climatic and behavioural conditions observed in the naturally ventilated Singapore buildings was approximately 2 K cooler than the neutrality of 28.5° C actually observed. Turning to field experiments in air conditioned buildings in the tropics, the PMV model's predicted neutralities for the present Singapore study, as well as those by Auliciems and de Dear (1986) and Busch (1990), were all slightly warmer than the empirically observed neutralities (by

circa 1 K). Comparing this positive one degree discrepancy between theory and observation with the negative discrepancy of approximately two degrees in the present naturally ventilated building study supports Humphreys (1976, 1981) and Auliciems' (1981) hypothesis that neutral temperatures shift towards the prevailing level of warmth in buildings. Busch arrived at a similar conclusion in his Bangkok study, noting that Thai office workers in naturally ventilated buildings '...expressed satisfaction with temperatures and humidities well above those deemed acceptable in the HVAC industry' (Busch 1990).

These field experiment comparisons bring us back to the general features of perception postulated in the introduction of this paper. Perception of indoor climate, at least as measured by a seven-point scale, appears not to be strictly determined by the indoor climatic and behavioural (clothing and metabolic) variables that control bodily heat balance. Instead, these data suggest that thermal perceptions are significantly attenuated by expectations. Given that climate-controlled office buildings have such constant temperatures and that thermal regimes in Singapore's naturally ventilated public housing are also relatively homogeneous, Singaporeans no doubt have well-established expectations of their indoor climates, and it is these expectations that appear to form the benchmarks for their thermal perceptions.

Taken out of context, the model of thermal perception discussed above raises the slightly absurd prospect of HVAC engineers being permitted by building occupants to shift indoor climatic design criteria arbitrarily. Field surveys in the US (Gagge and Nevins 1976; Elder and Tibbott 1981) demonstrated this to be far from the truth. They found widespread thermal dissatisfaction among occupants of office buildings in which air conditioning set-points were adjusted a few degrees in order to conserve energy. Carlton-Foss (1982) also found office building occupants to be 'barriers to the implementation of energy conservation strategies'. Apparently American office workers' expectations of their air conditioned work environments are not as malleable as energy/cost-conscious building service managers would like. In Bangkok, however, where air conditioning in offices may not yet be considered the norm, Busch (1990) found that at least 80% of workers in naturally ventilated buildings were quite tolerant of temperatures up to 30.5° C ET*, indicating that their indoor climatic expectations had not yet been raised to the levels of their more demanding counterparts in, for example, Singapore, Australia or the United States.

The third general feature of perception postulated in the introductory paragraphs of this paper implies that expectations are context specific, so atmospheric conditions that were demonstrated to be acceptable in naturally ventilated apartments in Singapore would have inevitably been met with widespread condemnation were they encountered in an office building in the same city. The contextual factors influencing expectations are no doubt legion, but near the top of the list probably are the questions of 'who is paying the energy bill?' and 'who is controlling the temperature?' These cognitive factors were also probably implicated in the surprisingly low

average dwelling temperature of 15.8° C observed by Hunt and Gidman (1982) in their nationwide UK field survey of 1000 houses. The average clothing insulation level of 0.83 clo observed in that study was well below the 2 clo level that would have been required to achieve a PMV of zero ('neutral' or 'just right'). Therefore it seems that the British have a greater tolerance of cold temperatures in their homes than is suggested by comfort standards. Similarly in Singapore, residents of naturally ventilated apartments have a greater tolerance of warm and humid indoor climates than suggested by the standards.

The foregoing is not intended to suggest that people in Singapore's public housing, or indeed in UK dwellings, would not opt for cooler temperatures (or warmer in the UK example), were all financial and other constraints to be waived. Even though household income seems certain to rise with the rapid development of Singapore's economy, the financial burden of air conditioning entire apartments is unlikely to disappear given the thermodynamic inefficiency of refrigeration technology. Moreover, even if air conditioning was to become economically more accessible, other considerations such as the adverse environmental impacts of excessive per capita fossil fuel consumption seem certain to increase the demand for energy conservation in buildings as we approach the greenhouse of the 21st century.

The psychological dimensions of thermal comfort discussed in this paper hold out opportunities in a time of renewed interest in passive solar buildings (Knudsen et al. 1989). It is a truism that passive buildings require active occupants. For example, in the behavioural sense occupants must adjust their clothing insulation, while in the psychological sense they must be prepared to adjust their expectations away from the traditional ideal of a homogeneous indoor climate. To date, the novelty of passive architecture in economically developed countries has probably selected out 'appropriate' occupants who are sympathetic to the low-energy design concept and quite willing to tolerate spatial and temporal perturbations of indoor climate. Therefore in economically developed countries, the challenge is to maintain that goodwill after the novelty has worn off and to broaden the acceptability of low-energy indoor climatic regimes to the general population as well as the enthusiasts. In newly industrializing countries such as Singapore, however, the main task is to resist escalating expectations of indoor climate despite rapidly increasing household incomes.

Conclusions

A field experiment in naturally ventilated apartments in Singapore found during day-time and early evening hours, a mean operative temperature of 29.6° C, mean RH of 74%, mean air velocity of 0.22 m/s, mean occupant clothing insulation level of 0.26 clo, and an average metabolic rate of 70 W/m². The mean thermal comfort vote on a seven-point scale was observed to be +0.66, which was about half-way between 'just right' and 'slightly warm'.

A field experiment in air conditioned office buildings in Singapore found a mean operative temperature of 23.5° C, mean RH of 56%, mean air velocity of 0.11 m/s, mean occupant clothing insulation level of 0.44 clo, and an average metabolic rate of 67 W/m². The mean thermal comfort vote on a seven-point scale was observed to be -0.34, which was on the cool margin of 'just right'.

The international and local standards of indoor climate and thermal comfort were not widely divergent from the thermal neutralities observed to date in field experiments in air conditioned buildings in the tropics. Observed neutralities were approximately one degree cooler than the various standards' prescriptions.

Thermal comfort responses in naturally ventilated apartments in hot humid climates were less well predicted by contemporary comfort theory and the standards. The neutral temperature empirically determined in the current field experiment was about 2 K above the prediction of the climate chamber-based PMV model.

These two Singapore studies indicate a discrepancy between thermal perception in naturally ventilated apartments and air conditioned offices of approximately 3 K which cannot be accounted for in terms of the basic heat balance variables (air and radiant temperatures, humidity, air velocity, clothing insulation and metabolic rate). This discrepancy does, however, seem consistent with a psycho-physiological model of thermal perception in which building occupants' indoor climatic expectations vary from one context to another.

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